

Investigations on Higher-Order Infinitary Logic

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Joint work with Olivier Hermant and Marc Aiguier



Extensions of First-Order Logic

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- ▶ Standard logic, but expressivity may be limited
- ▶ $\forall$ and $\exists$ abstract over individuals
- ▶ Example: Peano natural numbers (except the principle of induction)

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Higher-order and infinitary features have not been investigated together yet

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Has only been investigated for first-order geometric formulas

Define and investigate a higher-order infinitary logic

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- Define higher-order infinitary logic
- Study its semantics
- Extend Barr's theorem to it

Outline

Extension of Simple Type Theory

Higher-Order Infinitary Logic

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Representing Higher-Order (Infinitary) Logic

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Representing Higher-Order (Infinitary) Logic

- Higher-order logic: built using **simple type theory**
- Infinite conjunction $\bigwedge$ and infinite disjunction $\bigvee$
 - ▶ Take an infinite number of arguments
 - ▶ Not possible in simple type theory

■ Types and terms

$$\tau, \sigma ::= \iota \mid o \mid \tau \rightarrow \sigma \mid o_{\mathcal{I}}$$

$$t, u, A_i, F ::= x \mid c \mid t \ u \mid \lambda x. t \mid \{i \mapsto A_i\}_{\mathcal{I}} \mid \pi_j F$$

- ▶ Type $o_{\mathcal{I}}$ of families of propositions indexed by a countable set $\mathcal{I}$
- ▶ Family of propositions $\{i \mapsto A_i\}_{\mathcal{I}}$
- ▶ Projection $\pi_j F$ on the j -th component of F

Syntax and Typing

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■ Typing rules: usual ones, along with

- ▶ If $A_i : o$ for every $i \in \mathcal{I}$, then $\{i \mapsto A_i\}_{\mathcal{I}} : o_{\mathcal{I}}$
- ▶ If $F : o_{\mathcal{I}}$, then $\pi_j F : o$

Reduction Relation

■ **Reduction** $\hookrightarrow$ generated by

- ▶ $(\lambda x. t) u \hookrightarrow t[x \leftarrow u]$ (β -reduction)
- ▶ $\pi_j \{i \mapsto A_i\}_{\mathcal{I}} \hookrightarrow A_j$ (projection)
- ▶ $\lambda x. t x \hookrightarrow t$, where x is not free in t (η -contraction)
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■ The relation $\hookrightarrow_{\beta\pi}$ (without η -rules) is neither confluent nor terminating

- ▶ Each A_n reduces to B_n in n steps
- ▶ $\{n \mapsto A_n\}_{\mathbb{N}}$ does not reduce to $\{n \mapsto B_n\}_{\mathbb{N}}$ in a finite number of steps

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■ **Theorem:** Every well-typed term has a $\beta\pi$ -head normal form

- ▶ Adapting the reducibility technique [Tait 1967, Girard 1972]

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Logical Constants

■ Usual connectives

- ▶ $\perp$ of type o
- ▶ $\Rightarrow$ of type $o \rightarrow o \rightarrow o$
- ▶ $\forall_\tau$ and $\exists_\tau$ of type $(\tau \rightarrow o) \rightarrow o$
- ▶ $\neg A := A \Rightarrow \perp$

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■ Finite conjunction $\wedge$ and finite disjunction $\vee$

- ▶ $A \wedge B := \bigwedge_{\{0,1\}} \{0 \mapsto A, 1 \mapsto B\}$
- ▶ $A \vee B := \bigvee_{\{0,1\}} \{0 \mapsto A, 1 \mapsto B\}$

Example: Interactions between Researchers

- Researchers: terms of type ι

Relation between researchers: terms of type $\iota \rightarrow \iota \rightarrow o$

- ▶ Coauthors, colleagues, friends, ...

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$$(\exists R. R \ x \ y) \vee \bigvee_{\mathbb{N} \setminus \{0,1\}} \{n \mapsto \exists z_1 \dots \exists z_{n-1}. \exists R_1 \dots \exists R_n. R_1 \ x \ z_1 \wedge \dots \wedge R_n \ z_{n-1} \ y\}_{\mathbb{N} \setminus \{0,1\}}$$

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- Taking $y = \text{Erdős}$, generalization of Erdős problem

Natural Deduction Rules (1)

- Usual rules for $\perp, \Rightarrow, \forall, \exists$

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- **Computation**

- ▶ Conversion relation $\equiv$: reflexive, symmetric, transitive closure of $\hookrightarrow$
- ▶ Conversion rule

$$\frac{\Gamma \vdash A \quad A \equiv B}{\Gamma \vdash B} \text{ CONV}$$

Natural Deduction Rules (2)

■ Infinitary rules

$$\begin{array}{c} \frac{\Gamma \vdash \pi_i F \text{ for every } i \in \mathcal{I}}{\Gamma \vdash \bigwedge_{\mathcal{I}} F} \text{ ANDINF-I} \qquad \frac{\Gamma \vdash \bigwedge_{\mathcal{I}} F}{\Gamma \vdash \pi_j F} \text{ ANDINF-E} \\[2ex] \frac{\Gamma \vdash \pi_j F}{\Gamma \vdash \bigvee_{\mathcal{I}} F} \text{ ORINF-I} \qquad \frac{\Gamma \vdash \bigvee_{\mathcal{I}} F \quad \Gamma, \pi_i F \vdash C \text{ for every } i \in \mathcal{I}}{\Gamma \vdash C} \text{ ORINF-E} \end{array}$$

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■ Contexts Γ are finite

■ Derivation trees may have **infinite width**, but each branch has a **finite height**

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 - ▶ Leibniz's rule $\forall x. \forall y. \forall P. (P\ x \wedge x = y) \Rightarrow P\ y$

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 - ▶ Leibniz's rule $\forall x. \forall y. \forall P. (P\ x \wedge x = y) \Rightarrow P\ y$
- In higher-order infinitary logic, two equivalent principles:

$$\forall P. ((P\ 0 \wedge \forall x. (P\ x \Rightarrow P\ (s\ x))) \Rightarrow \forall x. P\ x)$$

$$\forall x. \bigvee_{\mathbb{N}} \{n \mapsto x = s^n\ 0\}_{\mathbb{N}}$$

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Intensional Models

- Higher-order (infinitary) logic is **not extensional** by default
 - ▶ We have $\vdash \perp \Leftrightarrow \perp \wedge \perp$ but we cannot prove $\vdash P \perp \Leftrightarrow P (\perp \wedge \perp)$
 - ▶ The interpretations of $P \perp$ and $P (\perp \wedge \perp)$ have to be different, so the interpretations of $\perp$ and $\perp \wedge \perp$ must be different

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 - ▶ The interpretations of $P \perp$ and $P (\perp \wedge \perp)$ have to be different, so the interpretations of $\perp$ and $\perp \wedge \perp$ must be different
- Sound notion of model
 - ▶ Based on **Heyting applicative structure** [Lipton-DeMarco 2005, H.-Lipton 2008]
 - ▶ Allows separating the domain of interpretation of formulas M_o from the Heyting algebra $\mathcal{H}$

Strong Version of Completeness

- Completeness $\approx$ exhibit a particular algebra based on provability
 - ▶ Intuitionistic logic $\rightarrow$ Heyting algebra, classical logic $\rightarrow$ Boolean algebra
 - ▶ Standard Lindenbaum algebra, which uses cuts

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- We construct a **cut-free Heyting/Boolean algebra** based on cut-free provability
- We build a specific model
 - ▶ Based on the **V-complex technique** [Prawitz 1968, Takahashi 67, Andrews 71]
 - ▶ Entangles syntax and semantics
 - ▶ Ensuring $M_o \neq \mathcal{H}$, with injection $\omega : M_o \rightarrow \mathcal{H}$

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- Completeness that entails the **cut-elimination theorem** for natural deduction

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- ▶ Geometric theory: all the axioms are geometric formulas

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■ Proof using **Friedman's translation**

- ▶ For first-order coherent formulas [Palmgren 2002, van den Berg 2019]
- ▶ For higher-order coherent formulas [T. 2024]

Friedman's Translation

- Applies $TA := (A \Rightarrow R) \Rightarrow R$ after universal quantifiers and implications:

$$\begin{aligned}x_T &:= x \\c_T &:= \begin{cases} \lambda p. \forall x. \textcolor{red}{T}(p\ x) & \text{if } c = \forall \\ \lambda p. \lambda q. p \Rightarrow \textcolor{red}{T}q & \text{if } c = \Rightarrow \\ c & \text{otherwise} \end{cases} \\(\lambda x. t)_T &:= \lambda x. t_T \\(t\ u)_T &:= t_T\ u_T \\(\{i \mapsto A_i\}_{\mathcal{I}})_T &:= \{i \mapsto (A_i)_T\}_{\mathcal{I}} \\(\pi_j\ F)_T &:= \pi_j\ F_T\end{aligned}$$

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- **Theorem:** If $\Gamma \vdash_{\text{CL}} A$ then $\Gamma_T \vdash_{\text{IL}} TA_T$

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- Choosing $R := A$, we get $\Gamma \vdash_{\text{IL}} A$

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Takeaway Message

- Extension of simple type theory with **families of propositions**
- **Higher-order infinitary logic**
 - ▶ Higher-order quantifiers $\forall$ and $\exists$
 - ▶ Infinite conjunction $\bigwedge$ and infinite disjunction $\bigvee$
- Results:
 - ▶ Semantics: soundness and completeness
 - ▶ **Cut elimination** for natural deduction
 - ▶ **Barr's theorem** for higher-order geometric formulas

- Investigate whether the relation $\hookrightarrow_{\beta\pi}$ can be modified to allow:
if $A_i \hookrightarrow_{\beta\pi}^* B_i$ then $\{i \mapsto A_i\}_{\mathcal{I}} \hookrightarrow_{\beta\pi} \{i \mapsto B_i\}_{\mathcal{I}}$

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Thank you!